

Reconstructed traversable wormholes and lensing features in linear $f(R, \mathcal{L}_m, T)$ gravity

Yohenba Soibam , Kangujam Priyokumar Singh  [Show more](#)
[Cite](#) [Add to Mendeley](#) [Share](#) | 10.1016/j.hedp.2026.101318

Access through your organi...

[Purchase PDF](#)

Article preview

Recommended articles

Abstract

In this work, we investigate static and spherically symmetric traversable wormhole geometries in the linear $f(R, \mathcal{L}_m, T) = R + \beta \mathcal{L}_m + \eta T$ gravity model. By adopting the Morris–Thorne metric with a finite redshift function $\Phi(r) = -\Phi_0 r_0 / (r + r_0)$, and an anisotropic fluid satisfying a radially varying equation of state $p_r(r) = \omega(r) \rho(r)$; $\omega(r) = \omega_0 + \omega_1 r_0 / r$, we reconstruct the shape function $b(r)$ from the modified field equations. The resulting reconstruction equation is a first-order linear differential equation, whose formal integral solution is obtained after imposing the throat condition $b(r_0) = r_0$. The flare-out condition, regularity of the reconstruction equation, and asymptotic flatness are used to identify an admissible parameter branch of the model. For representative parameter choices, the reconstructed solutions satisfy the required wormhole conditions. The radial null energy condition of the physical anisotropic matter sector is violated near the throat, as expected from the flare-out condition, while the tangential NEC remains non-negative over the representative numerical diagnostic domain. This indicates that the exoticity is mainly localized in the radial sector close to the throat. We also analyze the null geodesics and gravitational lensing properties of the reconstructed geometry. The off-throat circular null-orbit condition is governed by the redshift function and shows that no off-throat photon sphere exists for $0 < \Phi_0 < 4$. However, in the two-sided wormhole extension, the throat itself can act as an unstable light ring. Therefore, the finite deflection angles obtained for photons with closest approach radius $r_* > r_0$ describe the weak-to-moderate lensing branch away from the throat-light-ring limit. The embedding diagram confirms the regular flare-out structure of the reconstructed spatial geometry. These results show that the linear $f(R, \mathcal{L}_m, T)$ model admits reconstructed wormhole configurations with localized energy-condition violation and characteristic off-throat lensing features.

Keywords

Article preview

[Abstract](#)
[Introduction](#)
[Section snippets](#)
[References \(62\)](#)

Introduction

In General Relativity (GR) and its extensions, wormholes arise as nontrivial topological configurations connecting asymptotically distinct regions through a throat, with their physical viability depending on the relation between geometry and the supporting matter source. The concept originated with the Einstein–Rosen bridge [1] and earlier geometrical insights by Flamm [2], while Wheeler and collaborators later introduced the notion of multiply connected spacetime [3]. The modern framework of traversable wormholes was formulated by Morris and Thorne [4], who demonstrated that a static, spherically symmetric geometry described by a redshift function $\Phi(r)$ and a shape function $b(r)$ can, in principle, allow passage between two regions of spacetime. However, such configurations require the violation of classical energy conditions, particularly the null energy condition (NEC), at or near the throat [5], [6]. Furthermore, Olum [7] established that any form of superluminal travel necessarily involves negative energy densities, thereby linking traversable wormholes to exotic matter.

Within GR, this requirement poses a fundamental limitation. Known solutions such as the Ellis–Bronnikov wormhole [8], [9] rely on phantom scalar fields that violate the NEC, and extensive analyses have shown that ordinary classical matter cannot sustain wormhole geometries [6], [10]. This difficulty has motivated the study of modified theories of gravity, where additional geometric contributions may effectively mimic exotic matter and support wormhole configurations without requiring unphysical sources. Among the most extensively studied extensions is $f(R)$ gravity, in which the Einstein–Hilbert action is generalized to an arbitrary function of the Ricci scalar R . In this framework, higher-order curvature terms can support wormhole geometries while allowing the matter content to satisfy, or nearly satisfy, the energy conditions [11], [12], [13]. Recent investigations have further demonstrated the viability of wormholes in $f(R)$ gravity with additional physical ingredients such as Casimir effects and dark matter sources [14], [15]. Extensions such as $f(R, T)$ gravity [16], [17] and $f(R, \mathcal{L}_m)$ gravity [18] introduce explicit matter–geometry couplings, leading to modified field equations where effective stress–energy contributions arise from curvature. Several studies have shown that in such theories the violation of the NEC can be minimized or confined to localized regions near the throat [19], [20], [21]. More recent works have confirmed that viable wormhole solutions can be obtained in these frameworks under suitable parameter choices and matter distributions [22], [23], [24].

Beyond curvature-based modifications, wormholes have also been investigated in alternative gravitational frameworks. In Gauss–Bonnet and $f(\mathcal{G})$ gravity, higher-curvature invariants provide additional support for wormhole geometries [25], [26]. Teleparallel and symmetric teleparallel theories, such as $f(T)$ and $f(Q)$ [27], [28], [29], [30], as well as their extended matter–geometry coupled framework $f(Q, T)$ gravity [31], introduce torsion and non-metricity as fundamental geometrical

quantities and admit traversable wormhole solutions with physically acceptable matter sources. Similarly, scalar–tensor theories, including Brans–Dicke gravity, allow wormhole solutions sustained by scalar fields [32], [33], [34]. These results collectively indicate that modified gravity provides a consistent framework in which wormholes can exist with reduced or controlled exotic matter content. Recent studies have further broadened the range of viable wormhole constructions in modified gravity. These include investigations of traversable configurations in non-metric and $f(Q, T)$ gravity, Karmarkar-embedded geometries, charged wormholes, modified Gauss–Bonnet models, and other matter–geometry coupled scenarios [35], [36], [37], [38], [39], [40], [41], [42]. Complementary recent analyses have examined higher-dimensional, thin-shell, globally charged, and global-monopole wormhole configurations in several extended gravity frameworks [43], [44], [45], [46], [47], [48], [49]. Together, these developments emphasize the continuing importance of reconstruction techniques, energy-condition diagnostics, and geometrical constraints in assessing the physical viability of modified-gravity wormholes.

The $f(R, \mathcal{L}_m, T)$ gravity model proposed by Haghani and Harko [50] unifies the dependence on R , $\mathcal{L}_m$, and T within a single action, thereby extending both $f(R, T)$ and $f(R, \mathcal{L}_m)$ theories. The resulting field equations contain explicit matter–geometry coupling terms that modify the effective stress–energy sector. In the context of wormhole physics, previous studies have shown that such matter–geometry couplings can redistribute the effective sources entering the gravitational field equations, allowing the required energy-condition violation in some constructions to be reduced or partially attributed to the effective geometric sector [51]. Additional studies incorporating non-commutative effects and different equations of state suggest that wormhole solutions in this framework are not restricted to a particular matter prescription [52].

For the particular linear model considered in the present work, $f_R = 1$, so the curvature sector remains second order and contains no additional higher-derivative f_R contributions. The modification instead originates from the simultaneous algebraic dependence on $\mathcal{L}_m$ and T . Moreover, for the parameter branch considered here, Eq. (32) shows that the physical and effective null contractions are proportional. Consequently, the radial NEC violation required at the throat remains a genuine property of the physical anisotropic matter sector in the present construction. Thus, the model provides a comparatively transparent setting in which the effects of the two algebraic matter–geometry couplings can be studied in the reconstruction of the wormhole shape function.

From an observational perspective, gravitational lensing provides a powerful tool to probe compact objects and distinguish wormholes from black holes. Early work by Cramer et al. [53] suggested that wormholes could act as gravitational lenses with distinctive observational signatures. Subsequent studies analyzed lensing in specific wormhole spacetimes, including Ellis and Morris–Thorne geometries [54], [55], [56]. Depending on their metric functions, wormholes may exhibit lensing features different from those of black holes, including finite deflection angles in configurations without off-throat photon spheres [57], [58]. More recent developments have extended these analyses to strong-field regimes, including photon trajectories, shadows, and other lensing observables [59], [60], [61], [62]. These studies suggest that gravitational lensing may provide a useful avenue for probing wormhole geometries in astrophysical settings.

Although traversable wormholes have been widely investigated in modified gravity theories such as $f(R)$, $f(R, T)$, and $f(Q)$ gravity, many analyses proceed by prescribing one or both metric functions and then determining the associated matter content. Against this background, the present work focuses on a specific reconstruction route in which the shape function is not prescribed independently. We prescribe a regular non-constant redshift function and a radially varying radial equation of state and reconstruct $b(r)$ directly from the modified field equations. The resulting geometry is then subjected to throat and asymptotic viability tests, physical energy-condition diagnostics, an analysis of off-throat circular null orbits and the throat light ring, and a numerical evaluation of photon trajectories and deflection angles. This combined reconstruction and lensing analysis constitutes the specific focus of the present work.

The structure of this paper is as follows. In Section 2, we introduce the Morris–Thorne geometry, formulate the field equations of $f(R, \mathcal{L}_m, T)$ gravity, and specialize them to the linear model considered here. In Section 3, we prescribe the redshift function and radial equation of state, reconstruct the shape function, and establish the analytical and numerical viability conditions of the resulting wormhole branches. Section 4 examines the physical energy conditions and quantifies the localization of the NEC-violating matter. In Section 5, we analyze the null geodesics, off-throat circular-null-orbit condition, throat light ring, impact parameter, photon trajectories, deflection angle, and embedding geometry. The main results and limitations are summarized in the concluding section.

Organizational access

Get full-text access by signing in with your organisation

Access through **your organization**

Other access options

[Purchase PDF](#)

[Need help with access? ↗](#)

Section snippets

Wormhole geometry and field equations in $f(R, \mathcal{L}_m, T)$ gravity

We consider the static and spherically symmetric line element

$$ds^2 = -e^{\nu(r)} dt^2 + e^{\lambda(r)} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

where $\nu(r)$ and $\lambda(r)$ are functions of the radial coordinate r . For a Morris–Thorne

wormhole geometry, we write

$$\nu(r) = 2\Phi(r), \quad e^{\lambda(r)} = \left(1 - \frac{b(r)}{r}\right)^{-1},$$

so that

$$ds^2 = -e^{2\Phi(r)} dt^2 + \left(1 - \frac{b(r)}{r}\right)^{-1} dr^2 + r^2 d\Omega^2,$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. The function $\Phi(r)$ is the redshift function and $b(r)$ is the shape function. Traversability requires $\Phi(r)$ to remain finite everywhere in order to avoid horizons. The wormhole throat is located at $r \dots$

Reconstruction of the shape function

To close the system of field equations, we prescribe a radially varying equation of state relating the radial pressure and the energy density,

$$p_r(r) = \omega(r) \rho(r),$$

where

$$\omega(r) = \omega_0 + \omega_1 \frac{r_0}{r}.$$

Here ω_0 gives the asymptotic value of the equation-of-state parameter, while ω_1 measures the deviation from this value near the throat. This choice allows the supporting anisotropic matter distribution to vary with the radial coordinate.

For the redshift function, we adopt the finite non-constant form

$$\Phi(r) = -\frac{\Phi_0 r_0}{r + r_0},$$

...

Analytical energy-condition relations

Energy conditions provide useful diagnostic criteria for identifying the nature and localization of the matter source supporting a wormhole geometry. For an anisotropic fluid, the relevant combinations are written in terms of the energy density ρ , radial pressure p_r , and tangential pressure p_t . The null energy condition (NEC), weak energy condition (WEC), dominant energy condition (DEC), and strong energy condition (SEC) are respectively given by

$$\text{NEC:} \quad \rho + p_r \geq 0, \quad \rho + p_t \geq 0,$$

$$\text{WEC:} \quad \rho \geq 0, \quad \rho + p_r \geq 0, \quad \rho + p_t \geq 0,$$

$$\text{DEC:} \quad \rho \geq 0$$

...

Gravitational lensing analysis

Gravitational lensing encodes the behavior of null geodesics in curved spacetime and provides observationally relevant signatures such as photon trajectories, impact parameters, and deflection angles. In wormhole geometries, lensing also reveals whether the spacetime admits circular null orbits, throat light rings, or strong-deflection features. In this section, we analyze the lensing properties of the reconstructed wormhole geometry obtained in the previous sections.

Since the shape function $b(r)$...

Conclusion

The reconstruction developed here shows that an admissible class of static and spherically symmetric traversable wormholes can be obtained in the linear $f(R, \mathcal{L}_m, T) = R + \beta \mathcal{L}_m + \eta T$ model without prescribing the shape function independently. Once the finite redshift profile and the radially varying equation of state are specified, the modified field equations reduce to a first-order linear differential equation for $b(r)$. The resulting numerical branches satisfy the throat condition, the flare-out ...

CRedit authorship contribution statement

Yohenba Soibam: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Kangujam Priyokumar Singh:** Supervision, Investigation, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

References (62)

N. Godani

[Wormhole solutions in \$f\(R, T\)\$ gravity](#)

New Astron. (2022)

N. Kavya *et al.*

[Static traversable wormhole solutions in \$f\(R, \mathcal{L}_m\)\$ gravity](#)

Chinese J. Phys. (2023)

M.Z. Gul *et al.*

[Viable wormhole solutions in modified Gauss-Bonnet gravity](#)

Chinese J. Phys. (2024)

M. Tayde *et al.*

Impact of dark matter galactic halo models on wormhole geometry within $f(Q, T)$ gravity

Nuclear Phys. B (2024)

R. Saleem *et al.*

A study of wormhole solutions in $f(Q, T)$ gravity using some viable shape functions

Heliyon (2023)

A. Malik *et al.*

Effect of scalar potential and karmarkar condition on charged traversable wormholes

Phys. Lett. B (2026)

M.Z. Gul *et al.*

Viable wormhole solutions in modified Gauss–Bonnet gravity

Chinese J. Phys. (2024)

M.Z. Gul *et al.*

Impact of modified gravitational theory on the viability of wormhole structures

New Astron. (2024)

M.Z. Gul *et al.*

Study of viable charged wormhole solutions in $f(R, G)$ theory

New Astron. (2024)

E. Dai *et al.*

Imprints of monopole charge in morris–thorne type wormholes and energy conditions under the influence of higher dimensional gravity

Nuclear Phys. B (2025)



View more references

[View full text](#)

Recommended articles

Based on reading popularity

Short communication Open access

Cosmology in $f(R, L_m)$ gravity

Lakhan V. Jaybhaye, ..., P.K. Sahoo

Physics Letters B • Volume 831 • 2022 • Article 137148

Research article

Static traversable wormhole solutions in $\ell(\mathcal{R}, \mathcal{L}_m)$ gravity

N.S. Kavya, ..., S.V. Divya Rashmi

Chinese Journal of Physics • Volume 84 • 2023 • Pages 1-11

Research article

Accelerating scenarios of massive universe in $f(R, L_m)$ -gravity

Dinesh Chandra Maurya

New Astronomy • Volume 100 • 2023 • Article 101974

Research article

Baryogenesis in $f(R, L_m)$ gravity

Lakhan V. Jaybhave, ..., P.K. Sahoo

Physics of the Dark Universe • Volume 40 • 2023 • Article 101223

Research article

Casimir wormhole solutions in $f(R, \mathcal{L}_m)$ gravity

Mohan Khatri, J. Lalvohbika

Chinese Journal of Physics • Volume 89 • 2024 • Pages 1222-1235

Research article

Exploring the applicability of traversable wormhole formation in $f(\mathcal{R}, \mathcal{L}_m)$ gravity

Abdelghani Errehymy, ..., Abdel-Haleem Abdel-Aty

Chinese Journal of Physics • Volume 89 • 2024 • Pages 56-68

© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

